I argued AI was a cultural technology. *Here's what people actually do with it.*

Eunice Yiu · June 2026

In 2024, I argued that AI systems are best understood as a powerful new cultural technology, analogous to writing, print, libraries, and the internet. Rather than treating AI as an individual intelligent agent, my co-authors and I proposed that LLMs are best characterized as extraordinarily capable imitation engines: they aggregate and efficiently transmit the information already generated by humans, making cultural transmission cheap and effective at unprecedented scale.

The more consequential question today is how humans are actually interacting with AI, and whether AI is really being treated as a cultural technology. If AI is functioning as a cultural technology, we should see people using it as a thinking scaffold: asking questions, refining their own ideas, producing final responses that reflect their own thinking even when AI was consulted. If instead they are primarily delegating cognitive work wholesale, submitting AI's output with little or no modification, the economic and skill-formation implications are very different.

I ran a preregistered behavioral study to find out. The results complicate the cultural technology framing and point to a specific gap in how we currently measure AI's effects on work: platform logs and self-report surveys can capture whether someone used AI and how often, but not how much AI shaped what they produced.

STUDY DESIGN

Participants

227 adults recruited across the US in an ongoing preregistered study. Each was assigned to an open-ended task in the domain they reported using AI most frequently.

Design

Participants completed a survey on AI perceptions and usage, then were given a short open-ended task with an **optional AI assistant**. The choice about whether and how to use AI were left to the participant.

Task domains

Six domains drawn from Anthropic's Economic Index and prior work on how people use ChatGPT, covering the most common uses of AI:

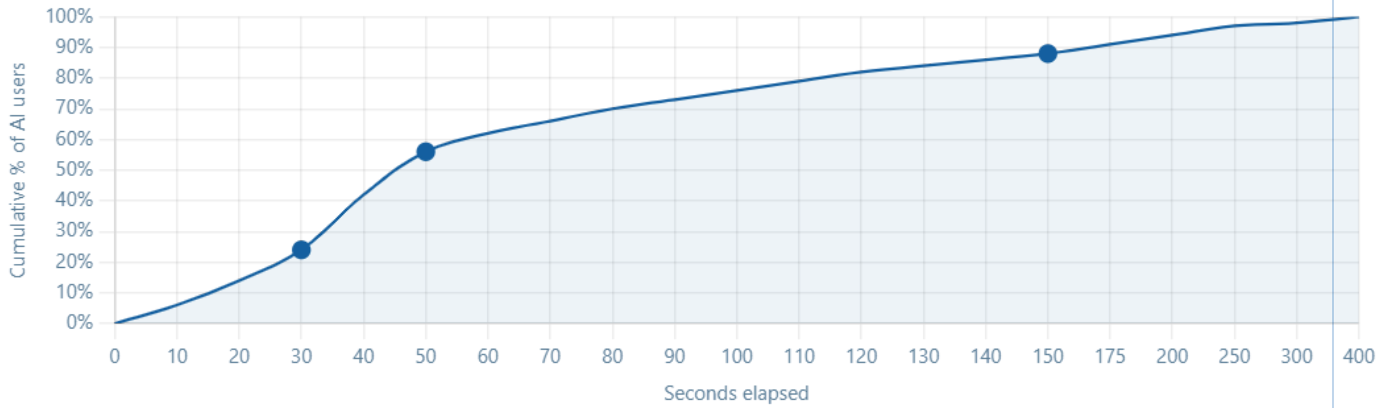
INFORMATION	Finding information and practical guidance, e.g., researching a topic, identifying options
WRITING	Writing, editing, or communicating, e.g., drafting a professional message or complaint
CREATION	Creative work: brainstorming, fiction, generating ideas, e.g., continuing a story, proposing new product concepts
LEARNING	Learning or understanding a topic in depth, e.g., how a scientific process works or a concept you have not encountered before
DECISION	Getting advice or talking through a decision, e.g., evaluating competing options or reasoning through a difficult choice
ANALYSIS	Summarizing content and forming opinions, e.g., critiquing an empirical study, interpreting findings, or evaluating limitations

FINDING 01

People consult AI immediately, even before composing their own answer

97% of AI users queried the AI assistant before typing in the response box. The median time to first AI click was 51 seconds, long enough to read the task, not long enough to think and write about it.

CUMULATIVE % OF AI USERS WHO HAD FIRST CONSULTED AI, BY SECONDS ELAPSED



All participants completed a task in the domain where they reported the highest AI use frequency.

FINDING 02

We coded each participant's AI query into one of three types:

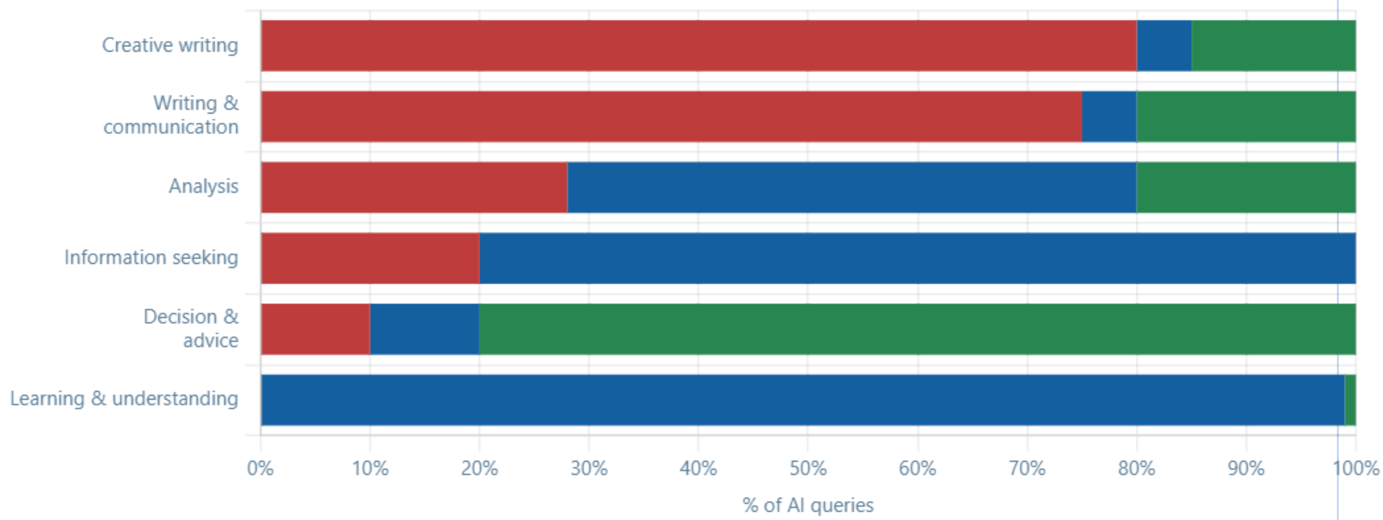
DELEGATE	Ask AI to produce the output directly. "Write a message about X." "Draft a response."
INFORM	Ask AI for information or explanation. "What is X?" "How does Y work?" "What are the pros and cons of Z?"
COLLABORATE	Share your own thinking and invite AI input, or ask AI to check your work. "I'm thinking X, what would you add?" "Is this reasoning sound?"

What the task calls for shapes how people query AI, and those queries are associated with how much AI appears in their final output

The way people queried AI varied substantially across task domains. The pattern is striking for writing tasks: whether for creative writing or practical communication, delegation is the dominant mode. People ask AI to produce the output and submit it. Learning tasks show the opposite: nearly all queries are INFORM, with people asking AI to explain or elaborate on a concept. Decision tasks are predominantly COLLABORATE, with people sharing their situation and reasoning together with AI.

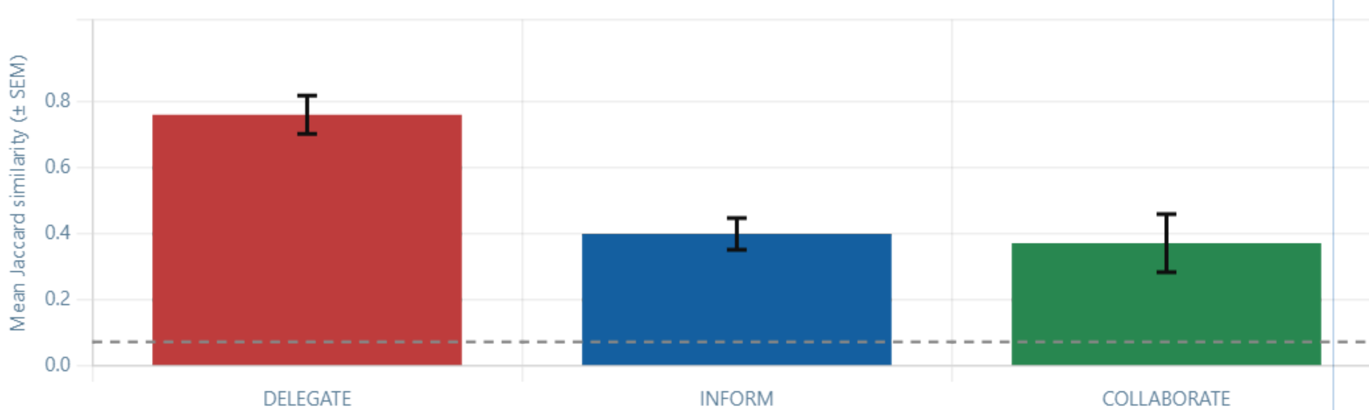
DELEGATE INFORM COLLABORATE

QUERY TYPE BY TASK DOMAIN, SORTED BY DELEGATION RATE



Query mode is also associated with how much AI output appears in the final response. We measure this with Jaccard similarity: the word-level overlap between what AI produced and what the participant submitted. A Jaccard of 1.0 means the participant's final text is word-for-word identical to AI's response; 0 means no shared words at all. For context, participants who did not consult AI at all had a mean Jaccard of 0.07 with the AI response. **DELEGATE queries produced a mean Jaccard of 0.76. INFORM and COLLABORATE queries produced 0.40 and 0.37.**

MEAN JACCARD SIMILARITY TO AI OUTPUT, BY QUERY TYPE (± SEM)



Error bars show ±1 SEM. Dotted line shows mean Jaccard for participants who did not consult AI ($M = 0.07$), the expected overlap from shared vocabulary alone. One-way ANOVA: $F = 8.08$, $p < .001$.

Task domain and query type together explain 21% of variance in lexical similarity ($R^2 = 0.210$). Task category alone explains 18.4% and query type alone 15.6%, with substantial overlap. For comparison, all person-level characteristics combined, including age, occupation, AI use tenure, and self-reported agency in AI use, explain 3.9%.

FINDING 03

What people say they do with AI doesn't match what they actually do

We classified each participant's stated AI usage into the same three categories as their actual query behavior. Only INFORM is well-calibrated: 59% of people who say they use AI to find information actually do.

The COLLABORATE group shows the sharpest gap. People who say they use AI collaboratively, sharing their thinking or checking their work, actually issue DELEGATE queries 42% of the time. **The group most committed to maintaining their own judgment is the one most likely to hand the task off when observed.**

STATED AI USAGE TYPE (COLUMN %) VS. REVEALED QUERY BEHAVIOR (ROW %)

Each row shows how people who described their usage in a given mode actually queried AI. Diagonal = aligned.

	DELEGATE	INFORM	COLLABORATE
DELEGATE	37%	40%	23%
INFORM	21%	59%	20%
COLLABORATE	42%	28%	30%

0% to 60% · Outlined diagonal = stated matches revealed

Furthermore, among the participants who copied AI output **verbatim** to complete their task, more than half had reported believing they were more **independent** than AI-reliant before the task.

Even when controlling for task type, the mismatch holds at the level of individual differences within task categories, suggesting it reflects genuine self-model error, not only task context.

FINDING 04

Participants showed meaningful awareness of AI uptake, but that awareness was not well calibrated

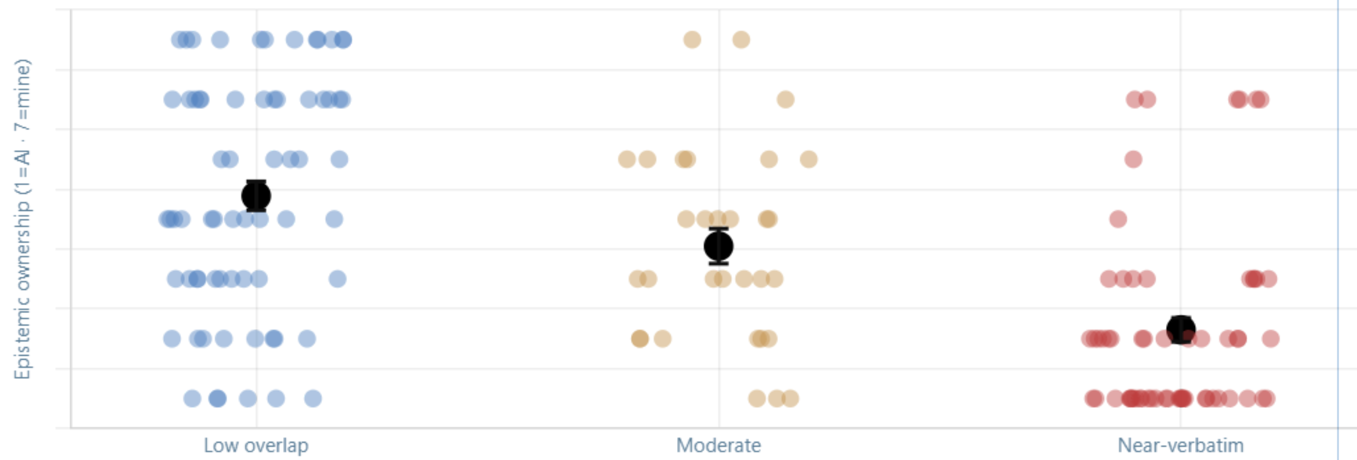
After completing their task, AI users rated how much the response reflected their own thinking versus the AI's (1 = mostly AI, 7 = mostly mine). These ratings tracked AI uptake in the expected direction: the more lexical overlap with AI output, the less ownership participants reported ($r = -0.51$, $R^2 = 0.26$, $p < .001$).

But sensitivity is not calibration. To assess calibration, we compared each participant's ownership rating to a simple benchmark: expected ownership = 7 - 6 × Jaccard, where verbatim copying implies ownership near 1 and no overlap implies ownership near 7. We then grouped responses by AI-output adoption: low overlap (Jaccard < .20), moderate overlap (.20-.79), and near-verbatim adoption (≥ .80).

Ownership declined across groups, but the range was compressed. Low-overlap users reported less ownership than their text overlap warranted, while near-verbatim users reported more. **The full range of AI influence was mapped onto a narrower range of perceived influence.** Among users whose similarity exceeded the non-AI baseline, 36% claimed more ownership than their similarity would predict.

EPISTEMIC OWNERSHIP BY DEGREE OF AI-OUTPUT ADOPTION

Ownership by adoption group



Error bars show ±1 SEM around group mean (filled circle)

People's miscalibration matters for how we study AI's effects on work. If people misestimate how much AI shaped what they produced, then self-reports about independence, effort, or authorship cannot tell us whether AI is augmenting human thinking or substituting for it. The downstream consequences for learning and skill formation remain an open question, but the measurement problem is already visible: ownership ratings, like platform logs and usage surveys, cannot be taken as direct evidence of cognitive engagement or output independence.

WHAT THIS MEANS

The measurement gap between AI use and AI influence

The cultural technology hypothesis describes one mode of engagement present in the data: INFORM and COLLABORATE queries, where people use AI as a thinking scaffold. But AI use is not one behavioral category. The same tool can support explanation, collaboration, or delegation depending on the task. In writing and creative tasks, delegation is common; in learning and decision tasks, people more often use AI to explain, elaborate, or reason with them.

Understanding AI's economic and societal effects requires measuring what AI actually contributes to what people produce, not just when it was accessed or how often. The findings here show that even within the domains where people use AI most, "AI use" is too coarse a unit of analysis. Some AI-assisted responses closely reproduced the model's output; others had little lexical overlap with it. For understanding AI's effects on work, those are not equivalent cases: they imply different degrees of model contribution to the final product. Ownership ratings compress this variation further, making the full range of AI influence partly invisible even to the people experiencing it.

Platform logs and self-report surveys, the current infrastructure for much AI economics research, cannot distinguish these cases on their own. Output-level behavioral measures can. Developing these measures, and using them to study the gap between AI use, AI influence, and perceived authorship, is a tractable empirical program with direct implications for how we measure augmentation, substitution, productivity, and skill change at scale. The central question for studying AI's effects on work is not simply whether the final text appears AI-generated, but how much of the work the model did: beginning with surface wording, and extending to argumentative structure, substantive ideas, and the shape of the final product.

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